When doing PQC inventorization and scanning data (documents, videos, images, etc.), what metadata should I look for when checking for security and encryption? For example, should I look at creation date, binary that created the file or encryption artefacts on the binary itself?

When conducting PQC inventorization and scanning files for cryptographic security, you should examine multiple layers of metadata and artifacts.[1] [2]

Key Metadata Categories to Examine

File-Level Metadata

- **Creation and modification timestamps:** Track when files were encrypted and last updated to identify data that requires long-term protection [1]
- **File format and type:** Document whether data is in transit, at rest, or in use [3]
- **Owner and application context:** Identify which system or application created/uses the file [4]
- **Data sensitivity classification:** Flag files containing high-value or long-lived secrets (medical records, national secrets, financial data) that need prioritized PQC migration [1]

Cryptographic Algorithm Metadata

- **Algorithm type and version:** Document specific cryptographic algorithms used (RSA, ECC, AES, etc.) with key lengths [3] [1]
- **Quantum vulnerability status:** Flag quantum-vulnerable algorithms (RSA, ECC) versus symmetric algorithms (AES) which face less urgent threats [1]
- **Cryptographic library and version:** Identify which libraries provide the encryption (OpenSSL, BoringSSL, etc.) [4] [3]
- **Protocol details:** For network data, capture TLS/SSH versions and cipher suites [5]

Binary and Executable Artifacts

When examining binaries that created or encrypted files:[6] [7]

- **Cryptographic API calls**: Static analysis to identify calls to crypto libraries and functions
- **Embedded certificates and keys**: Scan for hardcoded cryptographic materials
- **Signature verification keys**: Document signing algorithms used in the binary itself
- **Library dependencies**: Map which cryptographic modules are linked (e.g., FIPS-certified modules)[2]

Five-Pillar Discovery Framework

For comprehensive PQC inventory, scan across these pillars:[8]

1. **External Network**: TLS certificates, VPN endpoints, public-facing encryption
2. **Internal Network**: Internal TLS, SSH keys, IPSec tunnels
3. **IT Assets**: Endpoints, servers, HSMs, cloud KMS, containers
4. **Databases**: Encryption at rest, TDE configurations, connection encryption
5. **Code**: Source code repositories for hardcoded keys, crypto API usage

Critical Metadata Relationships

Document these relationships to build an effective Cryptographic Bill of Materials (CBOM):[2]

- Certificate → Signature key → Cryptographic module linkages
- Key metadata → Secret metadata connections
- Certificate metadata identifying verification keys
- Module metadata for FIPS-certified components

Risk Prioritization Metadata

Enhance your inventory with risk context tags:[4] [1]

- **Expiry status**: Flag certificates and keys nearing expiration
- **Policy compliance**: Mark non-compliant algorithms (SHA-1, weak keys)
- **Migration priority**: Tag items as "needs upgrade by 2025" or "quantum-vulnerable"
- **Remediation sequence**: Order by criticality and data sensitivity

Automated Discovery Tools

Manual tracking through questionnaires is insufficient. Use automated tools for:[1]

- Network traffic scanning (passive discovery of TLS/SSH parameters)
- Static code analysis (finding crypto API calls)
- Container image scanning (cryptographic libraries in containers)

- Cloud API integration (AWS KMS, Google Cloud KMS metadata extraction)[9] [4]

The inventory should capture rich contextual information including algorithm strength, usage patterns, key purpose, application dependencies, and expiration dates—enabling you to make informed migration decisions and maintain cryptographic agility.[9] [3]

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